

What is JavaScript?

1. A versatile, high-level programming language used to create interactive web content.
2. One of the three core technologies of the World Wide Web, alongside HTML and CSS.
3. Enables dynamic features like animated graphics, interactive forms, and real-time updates.
4. Supports multiple programming paradigms and is used for both client-side and server-side development.

What is Compiler

1. A computer program that translates computer code written in one programming language (the source language) into another language (the target language).
2. Primarily used for programs that translate source code from a high-level programming language to a lower-level language (e.g., machine code).
3. Performs complex processes like lexical analysis, preprocessing, parsing, semantic analysis, and code generation.
4. Essential for creating executable programs that can be run directly by a computer's hardware.

Comments in JavaScript

Method 1 : Single line comment

```
// This is a comment
```

Method 2 : Multiline comment

```
/*
```

```
1st line
```

```
2nd line
```

```
*/
```

Operators in JavaScript

`+` , `-` , `*` , `/`

Let `sum = a + b;`

// `a, b` : operands

// `+` : operators

Arithmetic Operators

`+` , `-` , `*` , `/`

1 Modulus `%`

2 Exponential `**`

3 Increment `++`

- `++a` preincrement, `a++` postincrement

4 Decrement `--`

Assignment Operators

`=` : Assign value

`\=` : Division Assignment (Often represented as `/=`)

`+=` : Addition Assignment

`-=` : Subtraction Assignment

`*=` : Multiplication Assignment

`%=` : Modulo Assignment

`**=` : Exponentiation Assignment

Comparison Operators

Comparison Operators

- Equal to: `==`
- Not equal to: `!=`
- Equal to and same type: `===`
- Not equal to or different type: `!==`
- Greater than: `>`
- Greater than or equal to: `>=`
- Less than: `<`
- Less than or equal to: `<=`

Logical Operators

Logical AND (&&): Checks if **both** conditions are true.

Example: `(age >= 18) && (hasID === true)`

Logical OR (||): Checks if **at least one** condition is true.

Example: `(isRaining === true) || (temperature < 10)`

Logical NOT (!): Reverses the boolean state of a condition.

Example: `!(isEmpty === true)`

Conditional Statements

if Statement

The **if statement** executes a block of code if a specified condition is true.

```
let color;  
if (mode == "dark mode") {  
  color = "black";  
}
```

if...else Statement

Conditional Logic

The **if...else** statement executes one block of code if a condition is true and another block if it is false.

```
if (mode == "dark mode") {  
    color = "black";  
} else {  
    color = "white";  
}
```

else if Statement

Use **else if** to specify a new condition if the first condition is false. The **else** block runs if none of the preceding conditions are true.

```
if (mode == "dark mode") {  
    color = "black";  
} else if (mode == "light mode") {  
    color = "white";  
} else {  
    color = "gray";  
}
```